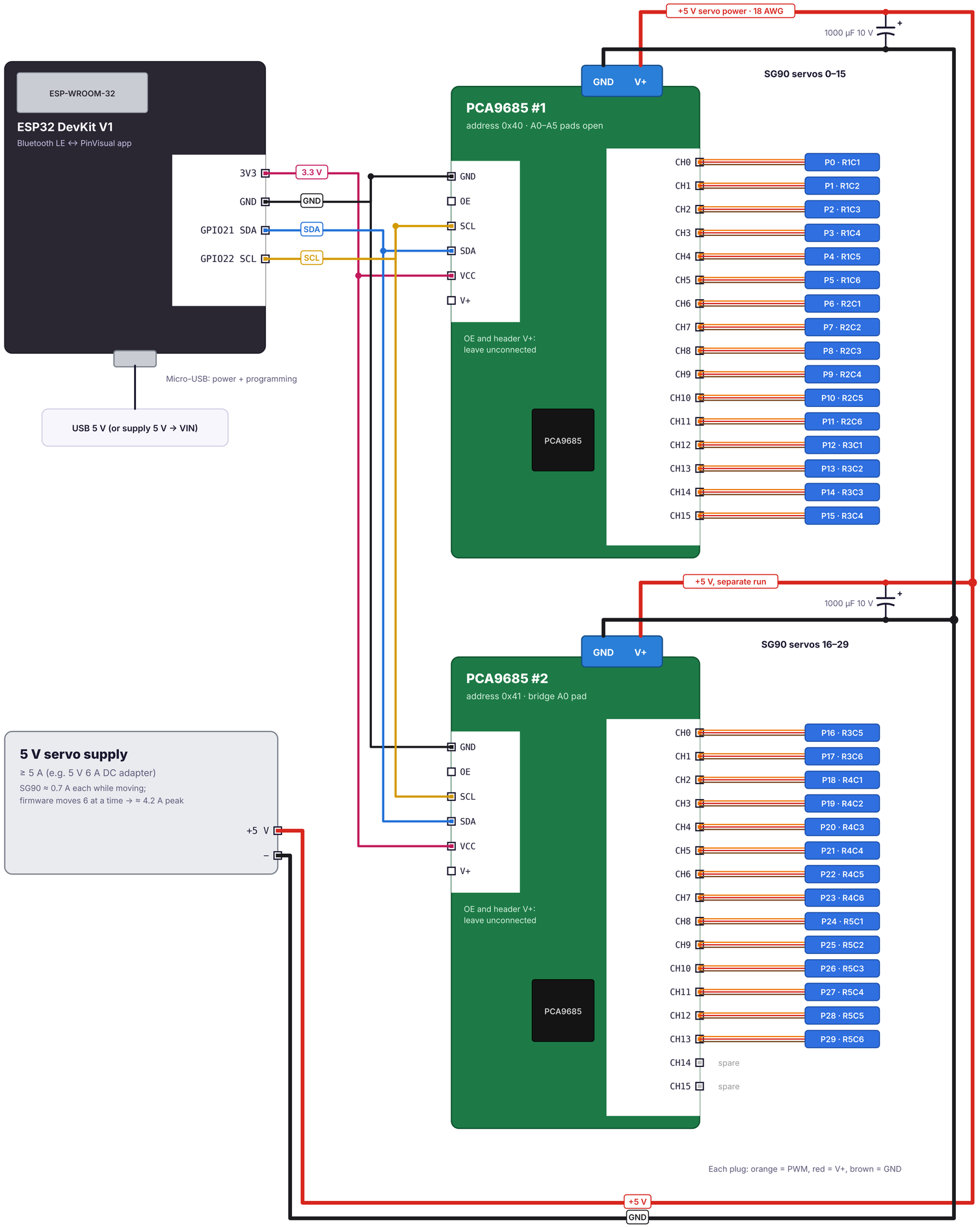
PinVisual — 30-servo circuit (5 × 6 display)

ESP32 drives two PCA9685 16-channel servo boards over I²C; the servos take power from the boards' V+ terminals. Firmware: PV_SERVO_BOARDS 1 · SDA 21 · SCL 22 · 0x40 / 0x41



Which servo is which pin (looking at the display from the user's side)

	C1	C2	C3	C4	C5	C6
R1	P0 0x40 · CH0	P1 0x40 · CH1	P2 0x40 · CH2	P3 0x40 · CH3	P4 0x40 · CH4	P5 0x40 · CH5
R2	P6 0x40 · CH6	P7 0x40 · CH7	P8 0x40 · CH8	P9 0x40 · CH9	P10 0x40 · CH10	P11 0x40 · CH11
R3	P12 0x40 · CH12	P13 0x40 · CH13	P14 0x40 · CH14	P15 0x40 · CH15	P16 0x41 · CH0	P17 0x41 · CH1
R4	P18 0x41 · CH2	P19 0x41 · CH3	P20 0x41 · CH4	P21 0x41 · CH5	P22 0x41 · CH6	P23 0x41 · CH7
R5	P24 0x41 · CH8	P25 0x41 · CH9	P26 0x41 · CH10	P27 0x41 · CH11	P28 0x41 · CH12	P29 0x41 · CH13

Board #1 (0x40): P0–P15

Board #2 (0x41): P16–P29, CH14–15 spare

Wire colours

- +5 V servo power
- GND (common)
- 3.3 V logic (VCC)
- I²C SDA (GPIO21)
- I²C SCL (GPIO22)
- Servo PWM

Rules

- Servo power goes only into each board's V+ screw terminal, never through the ESP32.
- Run separate +5 V / GND wires from the supply to each board (star wiring), 18 AWG.
- Every GND is common: supply –, both boards and the ESP32.
- VCC (board logic) comes from the ESP32's 3V3 pin, not from the 5 V supply.
- Board #2 needs its A0 address pad bridged with solder (0x41).
- Pin i → board i ÷ 16, channel i mod 16 (firmware/README.md).
- Board #2 takes the same 4 logic wires (I²C daisy-chain).
- Wires that cross without a dot are not connected.

Matches firmware/pinvisual-esp32/config.h and firmware/README.md · Planned wiring, not yet built on hardware